Constructing a low-rank implicit algorithm for solving high dimensional equations in hierarchical Tucker tensor format

Paolo Bosques-Paulet†, Dr. Joseph Nakao†, Dr. Gianluca Ceruti‡
† Swarthmore College, ‡ University of Innsbruck

Motivation

High dimensional PDEs are computationally expensive to solve and store. We can reduce the storage drastically by using a low-rank tensor decomposition. Test case:

$$u_t = u_{x_1x_1} + u_{x_3x_3},\ u = (x_1, x_2, x_3, x_4, t)$$

Heirarchical Tucker Decomposition

For full tensor solution for discretized Partial Differential Equation (PDE) we compute

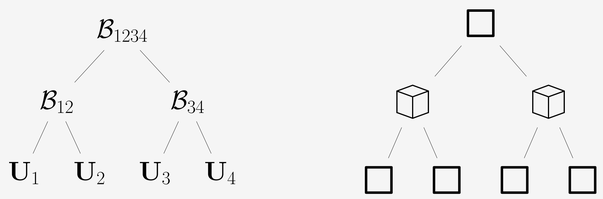


Table: Visualization of 4D HTD

$$\mathrm{vec}(\mathcal{U}) = (\mathbf{U_{34}} \otimes \mathbf{U_{12}})\,\mathbf{B_{1234}} \quad (1)$$

$$\mathrm{vec}(\mathcal{U}) = (\mathbf{U_4} \otimes \mathbf{U_3} \otimes \mathbf{U_2} \otimes \mathbf{U_1})\,(\mathbf{B_{34}} \otimes \mathbf{B_{12}})\,\mathbf{B_{1234}} \quad (2)$$

$$\mathbf{U_{34}} := (\mathbf{U_4} \otimes \mathbf{U_3})\,\mathbf{B_{34}}, \mathbf{U_{12}} := (\mathbf{U_2} \otimes \mathbf{U_1})\,\mathbf{B_{12}}.$$

$\mathbf{U_i}$ = Singleton leaf basis in the ith dimension
$\mathbf{B}_\alpha$ = Core tensor gluing bases in α_ℓ and α_r. $(\alpha = \alpha_\ell \cup \alpha_r)$

Storage savings: $\sim N^4 \rightarrow \sim (4Nr + 2r^3 + r^2)$

The low-rank algorithm in 4D

Step 1. Fully discretize PDE.
Step 2. Compute leaf bases and core tensors by HT Decomposition
Step 3. Solve K equations for $\mathbf{U}_i^{n+1}$
Step 4. Use Reduced Augmentation Procedure to extract $\hat{\mathbf{U}}_i^{n+1}$ from $\mathbf{K}_i$ and $\mathbf{U}_i^n$
Step 5. Use $\hat{\mathbf{U}}_\mathbf{i}^{n+1}$ to project fully discretized equation and solve for

$$\tilde{\mathbf{B}}_{12}^{n+1} = \mathbf{mat}(\mathcal{B}_{12}, a)\mathcal{B}_{1234}$$
$$\tilde{\mathbf{B}}_{34}^{n+1} = \mathbf{mat}(\mathcal{B}_{34}, a)\mathcal{B}_{1234}^T$$

and extract $\mathbf{B_{34}}$ and $\mathbf{B_{12}}$.
Step 6. Solve update equation for $\hat{\mathbf{B}}_\mathbf{1234}$
Step 7. Truncate $\mathrm{vec}\left(\mathcal{U}^{n+1}\right)$

HT-RAIL Equations

X^*: An orthogonal prediction matrix or flattened tensor used to project out dependence on a given dimension.
$\hat{X}$: A matrix or flattened tensor that contains information at time $n+1$ and time n that has not been truncated.
Example Eq in 12

$$\begin{aligned}\tilde{\mathcal{B}}_{12}^{n+1} \times_a \mathbf{U}_1^{n+1} \times_b \mathbf{U}_2^{n+1} \times \mathbf{U}_{34}^{n+1} = \\ \tilde{\mathcal{B}}_{12}^{n} \times_a \mathbf{U}_1^{n} \times_b \mathbf{U}_2^{n} \times_c \mathbf{U}_{34}^{n} + \\ \tilde{\mathcal{B}}_{12}^{n+1} \times_a \mathbf{D}_{x_1x_1} \mathbf{U}_1^{n+1} \times_b \mathbf{U}_2^{n+1} \times_c \mathbf{U}_{34}^{n+1} + \\ \tilde{\mathcal{B}}_{12}^{n+1} \times_a \mathbf{U}_1^{n+1} \times_b \mathbf{U}_2^{n+1} \times_c (\mathbf{I}_{r_4} \otimes \mathbf{D}_{x_3x_3}) \mathbf{U}_{34}^{n+1}\end{aligned}$$

K steps

$$\mathbf{K}_{1 \text{ or } 2}^{n+1} = \mathbf{U}_{1 \text{ or } 2}^{n+1} \mathbf{mat}\left(\tilde{\mathcal{B}}_{12}^{n}, 1 \text{ or } 2\right)\left(\mathbf{U}_{34}^{T,n+1}\mathbf{U}_{34}^{\star} \otimes \mathbf{U}_{2 \text{ or } 1}^{T,n+1}\mathbf{U}_{2 \text{ or } 1}^{\star}\right)$$

$$\mathbf{K}_{3 \text{ or } 4}^{n+1} = \mathbf{U}_{3 \text{ or } 4}^{n+1} \mathbf{mat}\left(\tilde{\mathcal{B}}_{34}^{n}, 3 \text{ or } 4\right)\left(\mathbf{U}_{12}^{T,n+1}\mathbf{U}_{12}^{\star} \otimes \mathbf{U}_{4 \text{ or } 3}^{T,n+1}\mathbf{U}_{4 \text{ or } 3}^{\star}\right)$$

B steps

$$\mathbf{vec}\hat{\tilde{\mathbf{B}}}_{12}^{n+1} = \left(\hat{\mathbf{U}}_{34}^T\mathbf{U}_{34}^{n+1} \otimes \hat{\mathbf{U}}_2^T\mathbf{U}_2^{n+1} \otimes \hat{\mathbf{U}}_1^T\mathbf{U}_1^{n+1}\right)\mathbf{vec}\tilde{\mathbf{B}}_{12}^{n+1}$$

$$\mathbf{vec}\hat{\tilde{\mathbf{B}}}_{34}^{n+1} = \left(\hat{\mathbf{U}}_{12}^T\mathbf{U}_{12}^{n+1} \otimes \hat{\mathbf{U}}_4^T\mathbf{U}_4^{n+1} \otimes \hat{\mathbf{U}}_3^T\mathbf{U}_3^{n+1}\right)\mathbf{vec}\tilde{\mathbf{B}}_{34}^{n+1}$$

$$\hat{\mathbf{B}}_{1234}^{n+1} = \left(\hat{\mathbf{U}}_2^T\mathbf{U}_2^{n+1} \otimes \hat{\mathbf{U}}_1^T\mathbf{U}_1^{n+1}\right)\hat{\tilde{\mathbf{B}}}_{12}^{n+1}\mathbf{B}_{1234}^{n}\hat{\tilde{\mathbf{B}}}_{34}^{n+1}\left(\mathbf{U}_3^{n+1}\hat{\mathbf{U}}_3^T \otimes \mathbf{U}_4^{n+1}\hat{\mathbf{U}}_4^T\right)$$

Reduced Augmentation

Question: How can we combine information from multiple time steps without rank blowup?
Answer:
Consider the bases $\mathbf{K}_i^{n+1}$ and $\mathbf{U}_i^n$ containing information at their respective time steps.
Factorizing:

$$[\mathbf{Q}_i, \mathbf{R}_i] = \mathrm{qr}([\mathbf{K}_i^{n+1}, \mathbf{U}_i^n])$$

Compressing using the SVD:

$$[\mathbf{u}_i, \boldsymbol{\sigma}_i, \mathbf{v}_i^T] = \mathrm{svd}(\mathbf{R}_i)$$

We can now consider only information corresponding to singular values above a specified tol.

$$\tilde{\mathbf{u}} = \mathbf{u}(i, 1:p)$$

Where p is the number of the smallest singular value greater than tol. We can define a reduced form of the augmented bases

$$\hat{\mathbf{U}}_i := \mathbf{Q}_i\tilde{\mathbf{u}}_i$$

Which preserves the most important information column space of $[\mathbf{K}_i^{n+1}, \mathbf{U}_i^n]$

Numerical tests

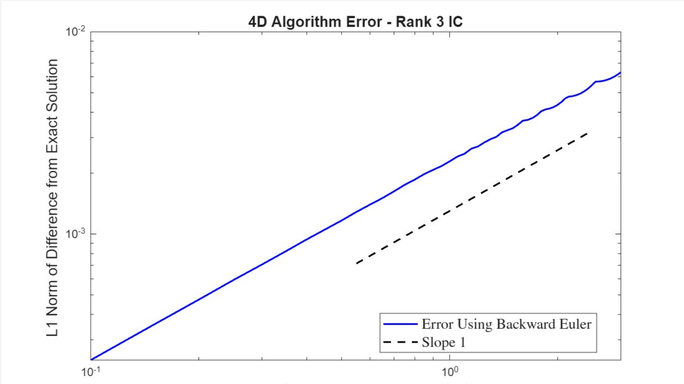


Figure: Error plot for Algorithm using Backward Euler

Initial condition:

$$u_0(x) = \sum_{k=1}^{3}\prod_{i=1}^{4}\sin\left(\frac{2k\pi}{L}x_i\right)$$

Ongoing and future work

1. High order accuracy
2. Extend algorithm to full 4D diffusion equation
3. Extend algorithm to advection-diffusion-reaction equations.
4. Apply algorithm to kinetic simulations (e.g., Fokker-Planck, Boltzmann-type equations)

References

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[2] Ceruti, G., Kusch, J. Lubich, C. A rank-adaptive robust integrator for dynamical low-rank approximation. *Bit Numer Math* 62, 1149–1174 (2022).
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